

1. Standing sound waves in a tube – an analog to a quantum mechanical particle in a box

Objective: For a simple tube, use an oscilloscope to compare the sound input by a speaker at one end to the sound received by a microphone at the other end.

Equipment Required:

TeachSpin Quantum Analog System: Controller, Sine wave generator capable of producing 1-50 kHz with a peak-to-peak voltage of 0.50 V
Two-Channel Oscilloscope

Setup:

Make a tube using the tube-pieces. Put the end-piece with the speaker on one end and the end- piece with the microphone on the other. Attach a BNC splitter to *SINE WAVE INPUT* on the Controller. Connect the output of your sine wave generator to one side of the splitter. Use a BNC cable to send the sound signal to the Channel 1 input of your oscilloscope. Plug the lead from the speaker end of your experimental tube to *SPEAKER OUTPUT* on the Controller. The same sine wave now goes to both the speaker and Channel 1. Connect the microphone output of the tube array to *MICROPHONE INPUT*. Connect *AC MONITOR* on the Controller to Channel 2 of the oscilloscope. Channel 2 will display the sound signal received by the microphone. Trigger the oscilloscope on Channel 1. Use the *ATTENUATOR* dial on the Controller to keep the signal on Channel 2 from going off scale.

Experiment:

Start at low frequency (100 Hz or less), and slowly increase the frequency.

Question:

What are you observing? How can you tell that you are at a resonance?

Did you notice the phase- shift when going through a resonance? (Note that, due to unknown phase shifts in the speaker, microphone, and electronics, the absolute phase between input and output channel can not be interpreted.)

Experiment:

Change the length of the tube and repeat the experiment.

Question:

Do the resonance frequencies change? Are they higher/lower when the tube is longer/shorter?

Take a full set of data for one tube length:

Measure and record the length of the tube. Measure the first 20 resonance frequencies. Assign the lowest resonance frequency the index number $n = 1$, and plot the resonance frequency f_n as function of its index number, n .

Background:

A resonance occurs when a standing sound wave has developed in the tube. The sound emitted by the speaker is reflected back and forth between the two hard end-walls of the tube. The resonance develops when, after a round trip in the tube, the sound wave is in phase with the wave emitted by the speaker. In this case, the emitted sound interferes with the reflected sound constructively. The condition for resonance is fulfilled when:

$$2L = nc/f = n\lambda$$

with the length of the tube L , the speed of sound c , the frequency f , the wavelength λ and an integer number $n=1,2,\dots\infty$. Resonances are observed when the tube length is an integer multiple of $\lambda/2$.

Analyze the data:

From the resonance frequencies plotted as function of their index n , you can calculate the speed of sound c . Make a linear fit for your data. Calculate c from the slope and determine the uncertainty of your measurement.

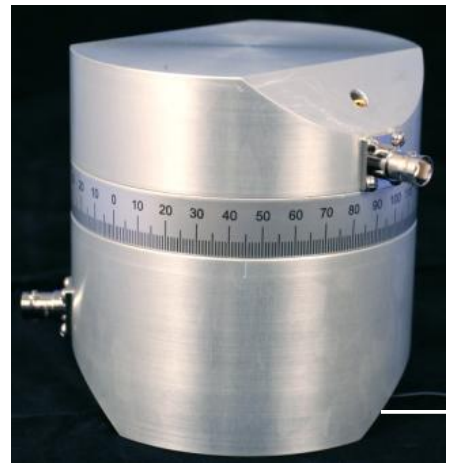
2. Modeling a hydrogen atom with a spherical resonator

2.1 Measure resonances in the spherical resonator and determine their quantum numbers

Objective: Determine the resonance frequencies for the spherical resonator and gather data to determine their angular-momentum quantum numbers.

Equipment Required: TeachSpin Quantum Analog System: Controller, Hemispheres, Sine wave generator capable of producing 1-50 kHz with a peak-to-peak voltage of 0.50 V Two-Channel Oscilloscope

Setup: Assemble two of the hemispheres so that the speaker is in the lower hemisphere and a microphone is in the upper hemisphere. (Looking carefully at the photo, you will see the speaker wire at the lower right.) Adjust the position of the upper hemisphere so that $\alpha = 180^\circ$ on the scale is at the reference mark. In this position, the speaker and the upper microphone are at opposite ends of a diameter. (The microphones will be one above the other.)



Atom Analog
Microphones are imbedded
under BNC connectors.
Speaker is at lower right.

Attach a BNC splitter of “tee” to SINE WAVE INPUT on the Controller. Connect the output of your sine wave generator to one side of the splitter. Use a BNC cable to send the sound signal to Channel 1 of the oscilloscope. Plug the lead from the speaker on the lower hemisphere to SPEAKER OUTPUT on the Controller. The same sine wave now goes to both the speaker and Channel 1.

Use a BNC cable to connect the microphone output from the upper hemisphere to MICROPHONE INPUT on the Controller. Connect AC MONITOR on the Controller to Channel 2 of the oscilloscope to display the sound signal received by the microphone. Trigger the oscilloscope on Channel 1. Use the ATTENUATOR dial on the Controller to keep the signal on Channel 2 from going off scale. Remember, with an attenuator, a higher reading on the dial gives a smaller signal.

Experiment:

Start at a low frequency and sweep the frequency up to about 8 kHz (8,000 Hz). Write down all the resonance frequencies you observe. (If you listen carefully, you may actually hear some of them.)

Objective:

Observe, qualitatively, the way the amplitude of the resonance signal depends on the location of the microphone.

Experiment:

We will now gather data that will allow us to infer the angular quantum numbers of the resonances. Go to the second resonance, at about 3680 Hz. Fine-tune the frequency until it is as close as possible to the peak of the resonance. Shift the curves on the oscilloscope horizontally so that a maximum of the microphone signal (Channel 2) is located in the center of the image and marked by a vertical line. Now, watching the signal on the oscilloscope, slowly rotate the upper hemisphere, with respect to the lower one, from $\alpha = 180^\circ$ to $\alpha = 0^\circ$.

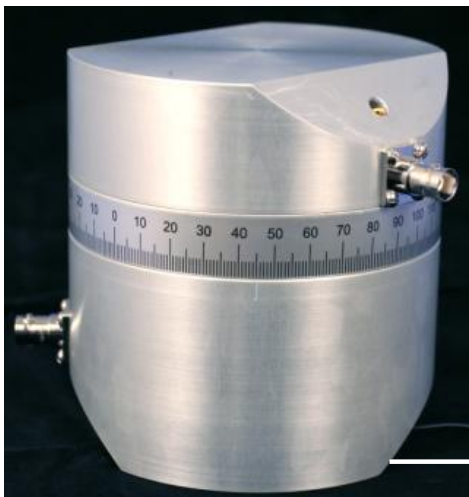
Questions:

How did the amplitude change? Did the signal change its sign? Determine the angle where the amplitude is zero. At which angles is the signal maximal? Do both extrema have the same amplitude?

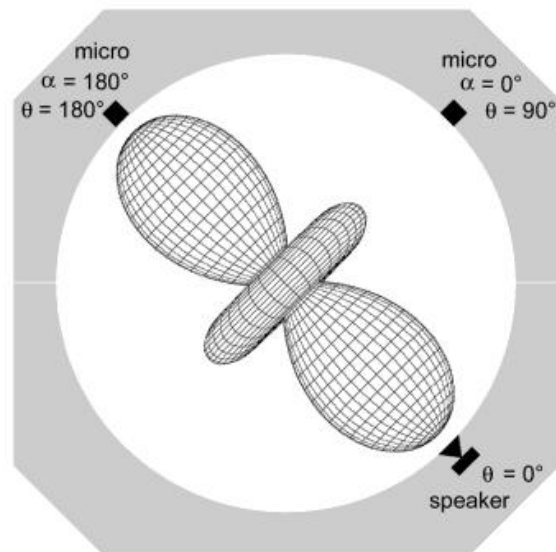
Note: Do not warm the aluminum parts too much by touching them with your hands. The speed of sound is temperature-dependent, and, in consequence, the resonance frequency would shift with temperature. While analyzing the angular dependence, the chosen generator frequency should remain on top of the resonance.

Analyze the data:

The angle α read on the scale is not a suitable angle for comparison with theory. Notice that the scale reading, α , running from $0 - 180^\circ$, tells you the rotation of the upper hemisphere about a vertical axis. The symmetry axis for this system, however, is determined by the speaker. The angle of interest, therefore, is measured using the speaker location as zero. To analyze the data, you must first use α to calculate the polar angle, θ . coordinates. To clarify how this works try the following.



Atom Analog
Speaker is at lower right



Sample Sound Amplitude Pattern

It is this angle which we use for polar coordinates. To clarify how this works try the following.

Assemble the sphere with the upper hemi- sphere set so that $\alpha = 180^\circ$. Temporarily open the resonator and notice that at this setting the speaker and the upper microphone are 180 degrees apart in space; $\theta = 180^\circ$. Reassemble the spheres, turn the upper sphere to $\alpha = 0^\circ$, and open the resonator again. You will see that the spatial separation of the speaker and microphone, the polar angle of interest, is now $\theta = 90^\circ$.

Both the speaker and microphone are at an angle of 45° with respect to the horizontal plane between the hemispheres. By rotating the hemispheres with respect to each other, the angle θ can be changed from $\theta = 90^\circ$ (at $\alpha = 0^\circ$) to $\theta = 180^\circ$ (at $\alpha = 180^\circ$). Intermediate angles can be calculated using the formula

$$\theta = \arccos\left(\frac{1}{2} \cos \alpha - \frac{1}{2}\right).$$

You have measured the θ -dependence of the spherical harmonic function $Y^m(\theta, \varphi)$ with $l=2$ and $m = 0$.